# Author Responses to Peer Reviews

Joint MECC 2026 – ASME JDSMC Submission #26

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We thank the Associate Editor and the reviewers for their constructive feedback and valuable suggestions on our manuscript. We have carefully revised the manuscript to address all the points raised. Below we provide responses to each of the specific comments and describe the corresponding revisions made in the paper.

| **Reviewers** | **Reviewer Feedback** | **Rebuttal Comments and Actions** |
|---|---|---|
| Associate editor, Reviewer 12, Reviewer 14 | Formatting errors and Writing issues. | 1. The writing issues were due to a missing cross-reference link while changing the paper format from the JDSMC template to the MECC template. These have been fixed.<br>2. The chemical symbols have been corrected from italic in math mode to roman format, as suggested by Reviewer 14. |
| Reviewer 18 | Independence of the saturated mode's dynamics from previous states. | The goal of the phrase was to convey that one mode (the saturated mode) of the switched dynamic model for NOx reduction is independent of initial conditions. This allows us to use the prediction error to classify data points as "close to saturation" when working out the statistics for aging detection.<br><br>We clarified this by adding more information to the referred sentences (in the first paragraph of Section 4). |
| Reviewer 14 | Interpretation of threshold $\beta$: Why is the threshold abruptly scaled to 250 for truck data? | The limited sample size causes $T_w$ to diverge from its asymptotic $\chi^2_\nu$ distribution under $\mathcal{H}_0$.<br><br>Added 2 paragraphs at the end of section-6 justifying the reason for threshold discrepancy. |

| **Reviewers** | **Reviewer Feedback** | **Rebuttal Comments and Actions** |
|---|---|---|
| Associate Editor, Reviewer 18, Reviewer 14 | • Explain the bottleneck of utilizing the limited data for diagnostics in the context of real-world applications.<br>• What operating conditions would ensure these necessary conditions are satisfied?<br>• How long is the system expected to wait for an OBD decision?<br>• Under driving conditions that do not trigger catalyst saturation, will the system generate false alarms? | Although only a fraction of the operating data is close to saturation (approximately 10% in real-world truck data), the convex parameter estimation framework utilizes the **entire** dataset. The non-saturated data points establish the lower boundary constraints ($\eta(k) \leq \eta_{\text{sat}}(k)$), which restrict the parameter space and prevent the estimates from collapsing.<br><br>We explicitly addressed the comments about dealing with the engineering limitations of the necessary conditions at the end of Section 6. |
| Reviewer 12 | Significance of the paper should be stated clearly | We separated the contributions into a dedicated paragraph at the end of the Introduction. |
| Reviewer 12 | The reader may want a clear comparison between experimental results and the model results. It is better to present figures than numbers. | Tables are utilized instead of time-series figures to convey the massive amount of detection results clearly and concisely. |
| Reviewer 12 | Literature review is not sufficient. Limitations of the existing works are not clear. | This paper focuses specifically on the narrow aspect of model-based SCR aging diagnostics under catalyst saturation. The current literature review and the included references justify the research objective, identify the gap, and map directly to this specific focus. We believe the existing references adequately represent the state-of-the-art for SCR aging diagnostics. |